Problem A. Beautiful Words

Time Limit	1000 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

You are given a string A of length N and a set S, containing M strings.

A cyclic permutation B_i of A, in which i is between 1 and N, is the string

$$B_i = A_i A_{i+1} \cdots A_{N-1} A_N A_1 A_2 \cdots A_{i-2} A_{i-1}$$

and its score is defined as the maximum length of a substring of B_i that is also a substring of some string in S.

A substring is defined as a contiguous sequence of letters. For example, `ab` and `dc` are substrings of `abfdc`, but `ad` and `fc` aren't substrings of `abfdc`.

Your task is to calculate the minimum score over all cyclic permutations of string A.

Input

The first line contains two positive integers N and M, ($1 \leq N \leq 10^5, 1 \leq M \leq 10^4$), representing the length of the string A and the size of the set S, respectively.

The second line contains the string A.

Each of the next M lines contains one string s_i, representing the i-th string in S.

All strings contain only lowercase English letters and it's guaranteed that the sum of lengths of all strings in S never exceeds 10^5 characters.

Output

Output an integer representing the minimum score over all cyclic permutations of string A.

Examples

Input	Output
7 3 acmicpc acm icpc maratona	3

Input	Output
11 4 competition oncom petition ztxvu fmwper	5

Input	Output
12 4 latinamerica zyvu okp wsgh kqpdb	0

Problem B. Creating Multiples

Time Limit	250 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

Malba is a very smart kid who loves to perform calculations. He has won several competitions, including the prestigious Tahan competition, in which he got the first prize, representing his country, Logonia.

He created a puzzle, in which he considers a number N, written in a certain base B, and represented by L digits. The objective of the game is to reduce at most one of the digits so that the new number, M, be a multiple of the number $B + 1$. But there is a catch: among the possible solutions you must choose one that renders M the smallest possible value.

For example, suppose that $B = 10$ and $N = 23456$. There are two ways in which M may be obtained: either we reduce the digit 4 to 0 or we reduce the digit 6 to 2. Thus, 4 must be changed to 0, hence $M = 23056$. Sometimes there is no solution, as is the case if $B = 10$ and $N = 102$. In this case, if we change the digit 1 to 9 we get a multiple of 11, but we are not allowed to increase the value of a digit!

Observe that it may be necessary to reduce the first digit to 0. For example, this is the case if $B = 10$ and $N = 322$.

Can you tell which digit should be reduced and what is its new value?

Input

The first line contains two integers B and L ($2 \leq B \leq 10^4, 1 \leq L \leq 2 \times 10^5$), representing the base and the number of digits of the number N, respectively.

The second line contains L integers $D_1, D_2, \ldots, D_L$ ($0 \leq D_i < B$ for $i = 1, 2, \ldots, L$), representing the digits of the number N. The first digit, D_1, is the most significant and the last, D_L, is the least significant.

Output

Output a line containing two integers, separated by one space. The first integer is the index of the digit to be changed (recall that the index of the first digit, D_1, is 1 and the index of the last digit, D_L, is L). The second integer is the new value of the digit. If there is no solution to the problem, output `-1 -1`. If N is already a multiple of $B + 1$ then output `0 0`.

Examples

Input	Output
10 5 2 3 4 5 6	3 0

Input	Output
10 3 1 0 2	-1 -1

Input	Output
2 5 1 0 1 1 1	4 0

Input	Output
17 5 3 0 0 0 0	1 0

Input	Output
16 4 15 0 13 10	1 14

Input	Output
16 5 1 15 0 13 10	0 0

Problem C. Escalator

Time Limit	500 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

You have just invented a new type of escalator: the double escalator. Regular escalators take people from one endpoint to another but not in the other direction, while the double escalator can take people from any one of its endpoints to the other one.

It takes 10 seconds for the double escalator to take a person from any of its endpoints to the other one. That is, if a person enters the double escalator from one of the endpoints at moment T, then they will leave at the other endpoint at moment $T + 10$ – this person won't be using the double escalator anymore at moment $T + 10$.

Any time that no one is using the double escalator, it will stop immediately. Thus, it is initially stopped.

When the double escalator is stopped and a person enters it from one of its endpoints, it will turn on automatically and move in the direction that this person wants to go.

If a person arrives at the double escalator and it is already moving in the direction that they want to go, they will enter it immediately. Otherwise, if it's moving in the opposite direction that they want to go, they will wait until it stops, and only then will they enter it. The escalator is so large that it can accommodate many people entering it at the same time.

The double escalator has a very weird effect, probably related to some quantum physics effect (or just chance): no person will ever arrive on the double escalator at the exact moment the escalator stops.

Now that you know how the double escalator works, you will be given the task of simulating it. Given the information about N people, including their time of arrival at the escalator and which direction they want to go, you have to figure out the last moment that the escalator stops.

Input

The first line contains one integer N ($1 \leq N \leq 10^4$), representing the number of people that will use the double escalator.

Each of the next N lines contains two integers t_i and d_i ($1 \leq t_i \leq 10^5, 0 \leq d_i \leq 1$), representing the time that the i-th person will arrive at the escalator and which direction they want to go. If d_i is equal to 0, they want to go from the left to the right endpoint, and if d_i is equal to 1, they want to go from the right to the left endpoint. All values of t_i are distinct and will be given in ascending order.

Output

Output one line containing the time that the last person will leave the double escalator.

Examples

Input	Output
3 5 0 8 0 13 0	23

Input	Output
3 5 0 7 1 9 0	29

Input	Output
3 5 0 10 1 16 0	35

Problem D. Getting in Shape

Time Limit	500 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

Juan decided to start working out and is willing to prepare a workout session.

He knows that some days he might not want to do all exercises from his workout session. He decided on some rules to avoid skipping the whole session and not exercising at all, while still allowing him to optionally skip some exercises.

The rules are:

- There will be only two types of exercises: A and B.
- After finishing an exercise of type B he moves to the next exercise, if there is one. Otherwise, the workout session ends.
- After finishing an exercise of type A there are two possibilities: he can move to the next exercise, or he can skip the next exercise and move to the one after that.
- The last exercise in a workout session must always be of type B.

Therefore, there might be different ways in which the workout session can be completed. For example, if the types of exercises in a workout session are `BAAB`, there are 3 ways in which the session can be completed: by doing all exercises, by skipping the 3rd one or by skipping the last one.

Juan wants to prepare his workout session in such a way that there are exactly N different ways in which the workout session can be completed. Can you help him?

Input

One positive integer N ($2 \leq N \leq 10^{15}$), representing the number of ways in which the workout session can be completed.

Output

Output a line containing a string, formed only with characters 'A' and 'B', representing the types of the exercises in the workout session. If there are multiple valid answers, output the lexicographically smallest answer. If there is no valid workout sessions, output a line containing the string "`IMPOSSIBLE`" (without quotes).

Examples

Input	Output
2	AB

Input	Output
4	ABAB

Input	Output
7	IMPOSSIBLE

Problem E. Handling the Blocks

Time Limit	500 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

A friend of yours invented a game and wants to know if you can solve it or if it's impossible.

He assembled a sequence of N blocks. Each block has a number engraved on it and some color. All numbers are distinct and between 1 and N, and different blocks can be of the same color.

The game works as follows: you can play as many turns as you want. In one turn, you choose two different blocks that share the same color and swap them.

You have to tell whether it is possible to get the entire sequence to be sorted into ascending order by numbers engraved on the blocks.

Input

The first line contains two integers N and K ($1 \le N \le 10^5$, $1 \le K \le N$), representing the number of blocks in the sequence and the number of different colors, respectively.

Each of the next N lines contains two integers n_i and c_i ($1 \le n_i \le N$, $1 \le c_i \le K$), representing the number and color of the i-th block, respectively.

Output

Output one line containing one character. If the sequence can be arranged in ascending order, write the upper case letter 'Y'; otherwise write the uppercase letter 'N'.

Examples

Input	Output
4 2 3 1 4 2 1 1 2 2	Y

Input	Output
4 2 2 1 4 2 1 1 3 2	N

Input	Output
3 1 1 1 2 1 3 1	Y

Problem F. Kathmandu

Time Limit 250 ms
Mem Limit 1048576 kB
OS Windows
Statement statements #1 (en) statements #2 (pt)

The pandemic is getting better and you can finally do the thing you've been dreaming of for the past few years: eat at your favorite restaurant! The restaurant happens to be in Kathmandu, but that's fine, you can always take a plane.

The problem is that planes almost always leave you restless. You consider yourself properly rested if you can sleep for T uninterrupted minutes, which means you are never awake from a certain moment t to $t + T$. Also, you're a very easy sleeper: you can fall asleep at the start of any minute and wake up at the end of any minute.

Of course, if you sleep too much you will miss all the airplane meals! That is unacceptable, as no opportunity for free food should go to waste.

Luckily, the airplane company sent you the whole flight schedule: the duration of the flight, D minutes, the number of meals that are going to be served, M, and the exact time they will serve the meals, y_i. You need to be awake at the time the meal is being served to be able to eat it, otherwise, the steward will not serve you. Since you're always hungry, you will devour the meal instantly.

Now you are wondering, for the optimal plane traveling experience, can you get properly rested and still eat all meals during the flight?

Input

The first line of input contains three integers, T, D, M ($1 \leq T, D \leq 10^5, 0 \leq M \leq 1000$), representing, respectively, the number of minutes you need to sleep without interruption to be properly rested, the duration of the flight and the number of meals that are going to be served during the flight.

Each of the next M lines contains an integer y_i ($0 \leq y_i \leq D$). These integers represent the times at which each meal is going to be served, and are given in chronological order.

Output

Output a line containing one character. If you can get properly rested and still eat all meals during the flight, write the upper case letter 'Y'; otherwise write the uppercase letter 'N'.

Examples

Input	Output
3 10 3 2 4 7	Y

Input	Output
4 10 3 2 4 7	N

Input	Output
5 5 0	Y

Input	Output
4 8 2 5 7	Y

Input	Output
4 8 2 3 4	Y

Problem G. Listing Passwords

Time Limit	3000 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

Michael is the manager of a barely known office and inside his room there is a locker with the money to pay the employees. Unfortunately, Michael forgot the password of the locker and it is now Dwight's responsibility to help his boss.

The password is a sequence of N digits, containing only zeroes and ones, and Michael remembers the value at some positions of the sequence, but not the whole password. Michael also remembers M intervals of the password that are palindromes – his mind memorizes palindromes, for some reason.

An interval is a palindrome if, and only if, the first and last digits of the interval are equal, the second and second to last digits are equal, and so on.

Now Dwight wants to know how hard it will be to recover the whole password. You can help Dwight by calculating the number of possible passwords that follow what Michael remembers.

Since the answer may be very large, output it modulo $10^9 + 7$.

Input

The first line contains the two integers N and M $(1 \leq N \leq 3 \times 10^5, 1 \leq M \leq 3 \times 10^5)$, representing how many digits the password has and the number of intervals Michael remembers as palindromes, respectively.

The second line contains N characters s_i, representing what digit Michael remembers about each position of the password. If s_i is '0' or '1', then the i-th digit of the password is 0 or 1, respectively. If s_i is '?', then Michael doesn't remember the i-th digit.

Each of the next M lines contains two integers l_i and r_i $(1 \leq l_i \leq r_i \leq N)$, which means that the interval of the password from the digit at position l_i to the digit at position r_i, inclusive, is a palindrome.

Output

Output the number of possible passwords that form a valid password modulo $10^9 + 7$. Since some of Michael's memories may be conflicting, in case there are no passwords that follow all his memories, print '0'.

Examples

Input	Output
5 2 1??0? 1 3 2 4	2

Input	Output
3 2 ??? 1 1 1 3	4

Input	Output
5 2 1???0 1 3 3 5	0

Problem H. Monarchy in Vertigo

Time Limit	250 ms
Mem Limit	1048576 kB
OS	Windows
Statement	statements #1 (en) statements #2 (pt)

Monarchy succession can be a tricky topic as it might take into account multiple factors such as descent, sex, legitimacy, and religion. Usually, the Crown is inherited by a sovereign's child or by a childless sovereign's nearest collateral line. Not really straightforward, right? And that's one of the reasons why, all over the world, monarchy is on the edge.

Nlogonia is still ruled by a monarchy though, luckily with simple succession rules. In general, there are only two aspects to take into account: "children comes before siblings" and "older comes before younger".

The reign servants maintain a beautiful and enormous tapestry where the bloodline of Constant, Nlogonia's first ruler, is drawn in the shape of a tree. Whenever a new family member is born a new branch from the parent to the child is drawn in the tapestry. This is such an important event that legend says that when Constant's descendants have a child they will never die before watching their child's name added to the tapestry. When someone dies, a cross is drawn close to that person's name. Whenever the current monarch dies, the tapestry is used by the servants to determine who the next ruler should be. In order to determine who that person is, the servants start from Constant and traverse the tree according to the rules described earlier, "children comes before siblings" and "older comes before younger". They descend the tree starting from Constant, followed by Constant's first child, followed by that child's first child, and so on, until reaching the first person alive or a family member that has no children left to follow, in which case they then go back to that person's parent and move into that parent's next child, repeating the process until the new monarch is determined.

After thousands of years in power Constant's bloodline is huge. To maintain the tapestry and, when the time comes, to determine who the next monarch should be are lengthy processes and Nlogonian servants decided it's time to modernize. They want to create a program to be used to maintain Constant's bloodline, which can also point who the next monarch should be after a ruler's tragic death. Given the importance of this task, the monarchy servants want to test the program by checking that it produces the correct output for all events that happened so far. There is one issue though, none of them is really good with programming and that's why they came after you for help.

More technically, each person in Constant's bloodline will be represented by a unique positive integer identifier. Whenever a new child is born, they take the next lowest unique identifier. Constant's identifier is equal to 1, and initially he's the only person alive. You will be given many events to process, in chronological order. Whenever someone dies, you should help the monarchy servants to figure out who the current monarch is. It's guaranteed that there will always be someone alive to rule.

Input

The first line contains an integer Q ($1 \leq Q \leq 10^5$), representing how many events should be processed. The following Q lines contain two integers t_i and x_i each, representing the i-th event type and argument. If t_i is equal to 1, then it means that person with identifier x_i had a new child. If t_i is equal to 2, then it means that person with identifier x_i died.

Output

For each event in which someone dies, you should print a line with an integer, representing the identifier of the current monarch.

Examples

Input	Output
8 1 1 1 1 1 2 2 1 2 4 1 2 2 2 2 5	2 2 5 3

Input	Output
4 1 1 1 1 2 2 2 1	1 3

Problem I. K-query

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

English	Vietnamese

Given a sequence of n numbers a_1, a_2 ... a_n and a number of k-queries. A k-query is a triple (i, j, k) ($1 \le i \le j \le n$). For each k-query (i, j, k), you have to return the number of elements greater than k in the subsequence a_i, a_{i+1} ... a_j.

Input

- Line 1: n ($1 \le n \le 30000$).
- Line 2: n numbers a_1, a_2 ... a_n ($1 \le a_i \le 10^9$).
- Line 3: q ($1 \le q \le 200000$), the number of k- queries.
- In the next q lines, each line contains 3 numbers i, j, k representing a k-query ($1 \le i \le j \le n$, $1 \le k \le 10^9$).

Output

- For each k-query (i, j, k), print the number of elements greater than k in the subsequence a_i, a_{i+1} ... a_j in a single line.

Example

Input	Output
5 5 1 2 3 4 3 2 4 1 4 4 4 1 5 2	2 0 3

Problem J. The day of the competitors

Time Limit	109 ms
Mem Limit	1572864 kB
Code Length Limit	10000 B
OS	Linux

The International Olympiad in Informatics is coming and the leaders of the Vietnamese Team have to choose the best contestants all over the country. Fortunately, the leaders could choose the members of the team among N very good contestants, numbered from 1 to N ($3 \le N \le 100000$). In order to select the best contestants the leaders organized three competitions. Each of the N contestants took part in all three competitions and there were no two contestants with equal results on any of the competitions. We say that contestant A is better than another contestant B when A is ranked before B in all of the competitions. A contestant A is said to be excellent if no other contestant is better than A. The leaders of the Vietnamese Team would like to know the number of excellent contestants.

Input

First line of the input contains an integer t ($1 \le t \le 10$), equal to the number of test cases. Then descriptions of t test cases follow. First line of description contains the number of competitors N . Each of the next N lines describes one competitor and contains integer numbers ai, bi, ci ($1 \le ai, bi, ci \le N$) separated by spaces, the order of i-th competitor's ranking in the first competition , the second competition and the third competition.

Output

For each test case in the input your program should output the number of excellent contestants in one line.

Note: Because the input is too large so we have 4 input files ~~and the total time limit is 4s (not 1s)~~.

Example

Input	Output
1 3 1 2 3 2 3 1 3 1 2	3

Problem K. Holes

Time Limit	1000 ms
Mem Limit	65536 kB
Input File	stdin
Output File	stdout

Little Petya likes to play a lot. Most of all he likes to play a game «Holes». This is a game for one person with following rules:

There are N holes located in a single row and numbered from left to right with numbers from 1 to N. Each hole has it's own power (hole number i has the power a_i). If you throw a ball into hole i it will immediately jump to hole $i + a_i$, then it will jump out of it and so on. If there is no hole with such number, the ball will just jump out of the row. On each of the M moves the player can perform one of two actions:

- Set the power of the hole a to value b.
- Throw a ball into the hole a and count the number of jumps of a ball before it jump out of the row and also write down the number of the hole from which it jumped out just before leaving the row.

Petya is not good at math, so, as you have already guessed, you are to perform all computations.

Input

The first line contains two integers N and M ($1 \le N \le 10^5$, $1 \le M \le 10^5$) — the number of holes in a row and the number of moves. The second line contains N positive integers not exceeding N — initial values of holes power. The following M lines describe moves made by Petya. Each of these line can be one of the two types:

- 0 a b
- 1 a

Type 0 means that it is required to set the power of hole a to b, and type 1 means that it is required to throw a ball into the a-th hole. Numbers a and b are positive integers do not exceeding N.

Output

For each move of the type 1 output two space-separated numbers on a separate line — the number of the last hole the ball visited before leaving the row and the number of jumps it made.

Examples

Input	Output
8 5 1 1 1 1 1 2 8 2 1 1 0 1 3 1 1 0 3 4 1 2	8 7 8 5 7 3

Problem L. Submerging Islands

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Vice City is built over a group of islands, with bridges connecting them. As anyone in Vice City knows, the biggest fear of vice-citiers is that some day the islands will submerge. The big problem with this is that once the islands submerge, some of the other islands could get disconnected. You have been hired by the mayor of Vice city to tell him how many islands, when submerged, will disconnect parts of Vice City. You should know that initially all the islands of the city are connected.

Input

The input will consist of a series of test cases. Each test case will start with the number N ($1 \le N \le 10^4$) of islands, and the number M of bridges ($1 \le M \le 10^5$). Following there will be M lines each describing a bridge. Each of these M lines will contain two integers Ui, Vi ($1 \le Ui, Vi \le N$), indicating that there is a bridge connecting islands Ui and Vi. The input ends with a case where N = M = 0.

Output

For each case on the input you must print a line indicating the number of islands that, when submerged, will disconnect parts of the city.

Example

Input	Output
3 3 1 2 2 3 1 3 6 8 1 3 6 1 6 3 4 1 6 4 5 2 3 2 3 5 0 0	0 1

Problem M. Distinct Colors

Time Limit 1000 ms
Mem Limit 524288 kB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \ldots, n$, and node 1 is the root. Each node has a color.

Your task is to determine for each node the number of distinct colors in the subtree of the node.

Input

The first input line contains an integer n: the number of nodes. The nodes are numbered $1, 2, \ldots, n$.

The next line consists of n integers $c_1, c_2, \ldots, c_n$: the color of each node.

Then there are $n-1$ lines describing the edges. Each line contains two integers a and b: there is an edge between nodes a and b.

Output

Print n integers: for each node $1, 2, \ldots, n$, the number of distinct colors.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c_i \leq 10^9$

Example

Input	Output
5 2 3 2 2 1 1 2 1 3 3 4 3 5	3 1 2 1 1

The Queen of Nlogonia is a fan of mazes, and therefore the queendom's architects built several mazes around the Queen's palace. Every maze built for the Queen is made of rooms connected by corridors. Each corridor connects a different pair of distinct rooms and can be transversed in both directions.

The Queen loves to stroll through a maze's rooms and corridors in the late afternoon. Her servants choose a different challenge for every day, that consists of finding a simple path from a start room to an end room in a maze. A simple path is a sequence of *distinct* rooms such that each pair of consecutive rooms in the sequence is connected by a corridor. In this case the first room of the sequence must be the start room, and the last room of the sequence must be the end room. The Queen thinks that a challenge is good when, among the routes from the start room to the end room, *exactly* one of them is a simple path. Can you help the Queen's servants to choose a challenge that pleases the Queen?

For doing so, write a program that given the description of a maze and a list of queries defining the start and end rooms, determines for each query whether that choice of rooms is a good challenge or not.

Input

Each test case is described using several lines. The first line contains three integers R, C and Q representing respectively the number of rooms in a maze ($2 \leq R \leq 10^4$), the number of corridors ($1 \leq C \leq 10^5$), and the number of queries ($1 \leq Q \leq 1000$). Rooms are identified by different integers from 1 to R. Each of the next C lines describes a corridor using two distinct integers A and B, indicating that there is a corridor connecting rooms A and B ($1 \leq A < B \leq R$). After that, each of the next Q lines describes a query using two distinct integers S and T indicating respectively the start and end rooms of a challenge ($1 \leq S < T \leq R$). You may assume that within each test case there is at most one corridor connecting each pair of rooms, and no two queries are the same.

The last test case is followed by a line containing three zeros.

Output

For each test case output $Q + 1$ lines. In the i-th line write the answer to the i-th query. If the rooms make a good challenge, then write the character 'Y' (uppercase). Otherwise write the character 'N' (uppercase). Print a line containing a single character '-' (hyphen) after each test case.

Sample Input

```
6 5 3
1 2
2 3
2 4
2 5
4 5
1 3
1 5
2 6
4 2 3
1 2
2 3
1 4
1 3
1 2
0 0 0
```

Sample Output

```
Y
N
N
-
N
Y
Y
-
```

Problem O. Proving Equivalences

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Consider the following exercise, found in a generic linear algebra textbook. Let A be an $n \times n$ matrix. Prove that the following statements are equivalent:

- (a) A is invertible.
- (b) $Ax = b$ has exactly one solution for every $n \times 1$ matrix b.
- (c) $Ax = b$ is consistent for every $n \times 1$ matrix b.
- (d) $Ax = 0$ has only the trivial solution $x = 0$.

The typical way to solve such an exercise is to show a series of implications. For instance, one can proceed by showing that (a) implies (b), that (b) implies (c), that (c) implies (d),and finally that (d) implies (a). These four implications show that the four statements are equivalent. Another way would be to show that (a) is equivalent to (b) (by proving that (a) implies (b) and that (b) implies (a)), that (b) is equivalent to (c), and that (c) is equivalent to (d).

However, this way requires proving six implications, which is clearly a lot more work than just proving four implications! I have been given some similar tasks, and have already started proving some implications. Now I wonder, how many more implications do I have to prove? Can you help me determine this ?

Input

On the first line one positive number: the number of testcases, at most 100. After that per testcase:

- One line containing two integers n ($1 \le n \le 20\ 000$) and m ($0 \le m \le 50\ 000$): the number of statements and the number of implications that have already been proved.
- m lines with two integers s1 and s2 ($1 \le s1, s2 \le n$ and $s1 = s2$) each, indicating that it has been proved that statement s1 implies statement s2.

Output

Per testcase:
One line with the minimum number of additional implications that need to be proved in order to prove that all statements are equivalent.

Example

Input	Output
2 4 0 3 2 1 2 1 3	4 2